

PRAJNA

Physics-Informed Neural Network for
Autonomous Reactor Safety Advisory

Haryana Ideathon 2026

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THE PROBLEM

Critical infrastructure faces two failure modes existing systems can't handle fast enough



Fast Thermal-Hydraulic Transients

LOCA · Reactivity-Initiated Accidents · Steam Generator Tube Rupture · Station Blackout — emergencies that unfold in seconds, not minutes.



Sensor Tampering & Decalibration

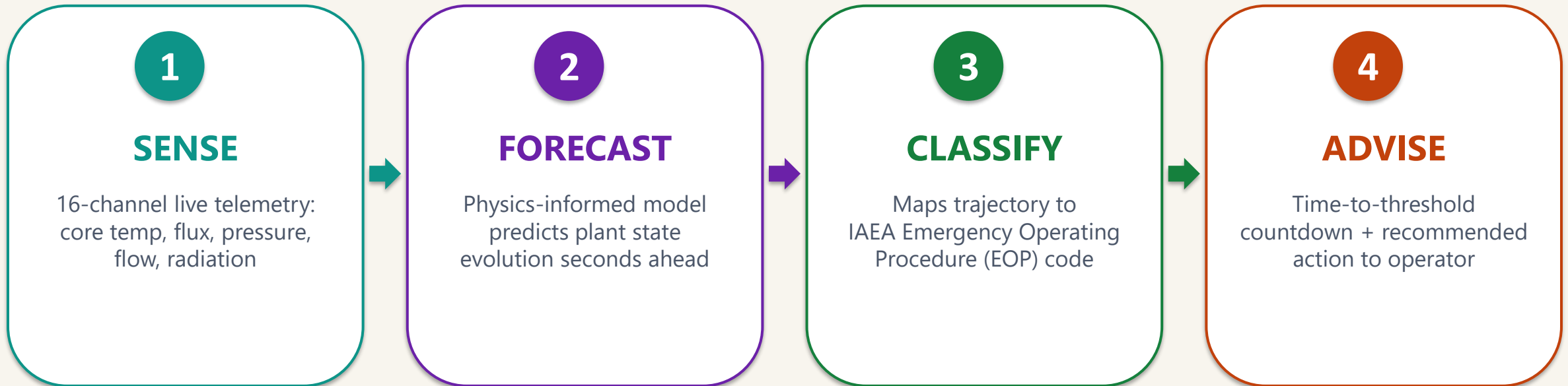
Falsified or drifting sensor data can silently mislead both human operators and standard monitoring software.

WHY EXISTING SYSTEMS FALL SHORT

Physics simulators (RELAP5, TRACE): highly accurate, but take seconds to minutes per timestep — too slow for a live emergency. Pure ML models: fast, but unconstrained by physics — their errors aren't bounded, so operators can't trust them in safety-critical loops.

THE SOLUTION

A physics-grounded AI layer that watches, predicts, and advises — without ever touching the controls



✓ **Zero-actuation by design — PRAJNA advises the human operator. It never issues a command to a physical control system.**

ARCHITECTURE

Two-tier design: a large model that learns, a tiny model that runs

FOUNDATION MODEL · TRAINING

Mamba-2 selective state-space backbone +
Fourier Neural Operator (3D core field resolution)

264.7M parameters · 1.01 GB checkpoint

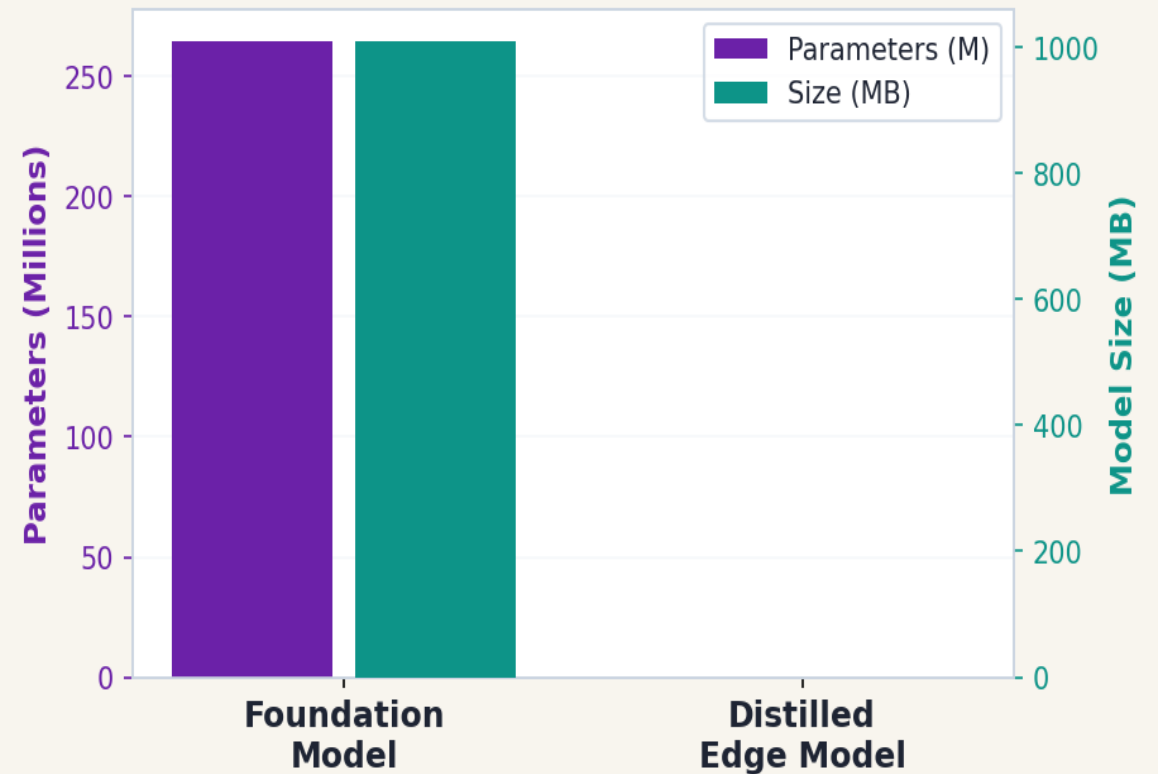
↓ knowledge
distillation

DISTILLED EDGE MODEL · DEPLOYMENT

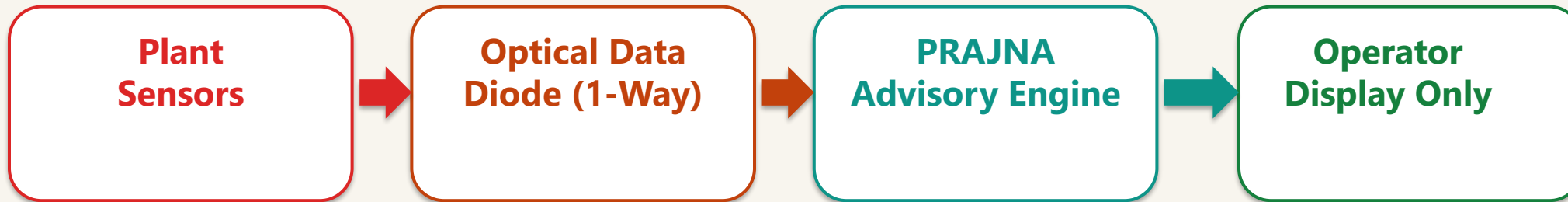
INT8-quantized ONNX graph — runs on CPU, no GPU
required

30,577 parameters · 47.16 KB

Model Compression: 8,660x Fewer Parameters



A read-only advisory system, physically incapable of being turned into an attack vector



Data flows strictly left to right. PRAJNA has no physical or logical path back to actuators — hardware-enforced.

Hardware Root of Trust

TPM 2.0 measured boot verifies system integrity before PRAJNA is trusted to advise — tampering with the software stack is detectable before the system goes live.

Physics as a Tamper Detector

Sensor data that violates energy-balance physics spikes the model's residual loss — flagging decalibration or falsified telemetry automatically.

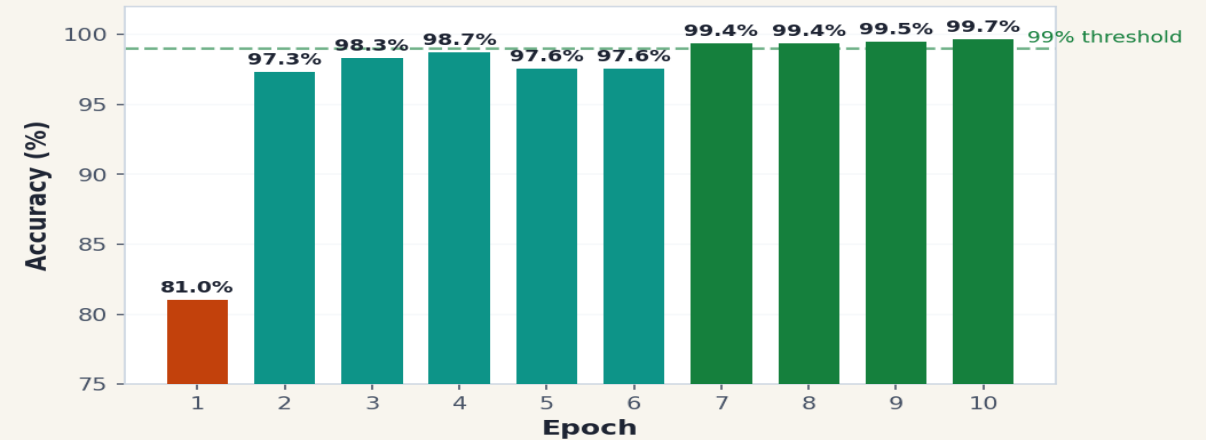
TRAINING RESULTS

10-Epoch Training on CUDA GPU — 264.7M Parameter Foundation Model

Training & Validation Loss Convergence



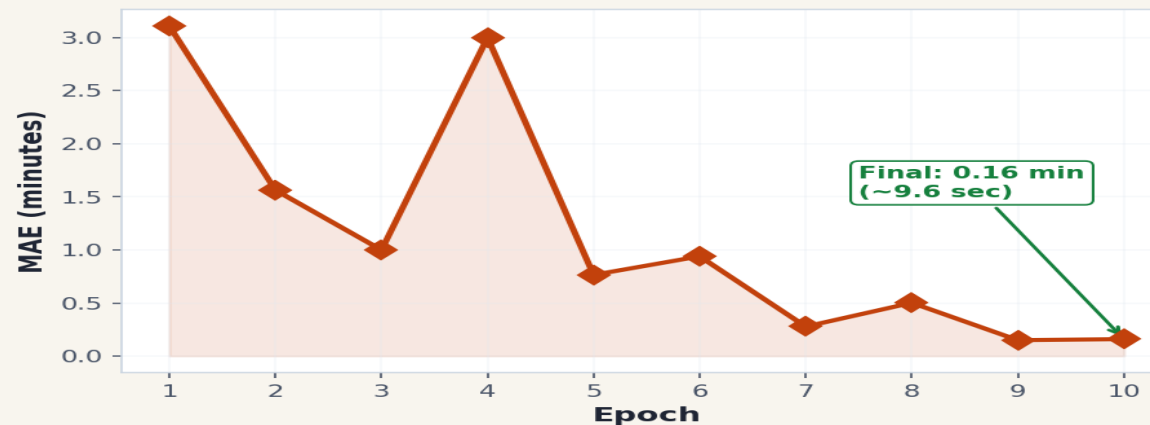
EOP Classification Accuracy per Epoch



KEY RESULTS

- ✓ 99.65% Accuracy
- ✓ 0.27 Final Loss
- ✓ ~9.6s MAE
- ✓ 2.3h Training

Delay Prediction MAE Over Training



VERIFIED RESULTS

Measured on-disk and via live benchmark — not projected

47.16 KB

Edge model size (INT8 ONNX)

≈ 44 μs

P50 CPU inference latency

22,000+

Inferences/sec throughput

$R^2 = 0.93$

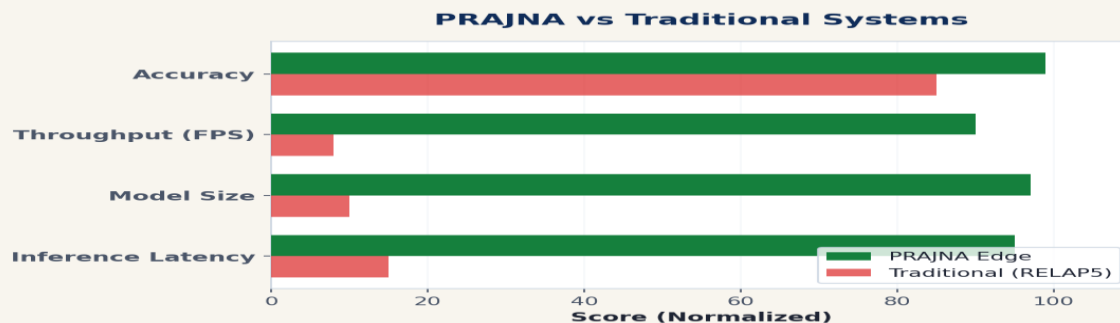
Regression fit, 16 channels

99.0%

EOP classification accuracy

6 / 6

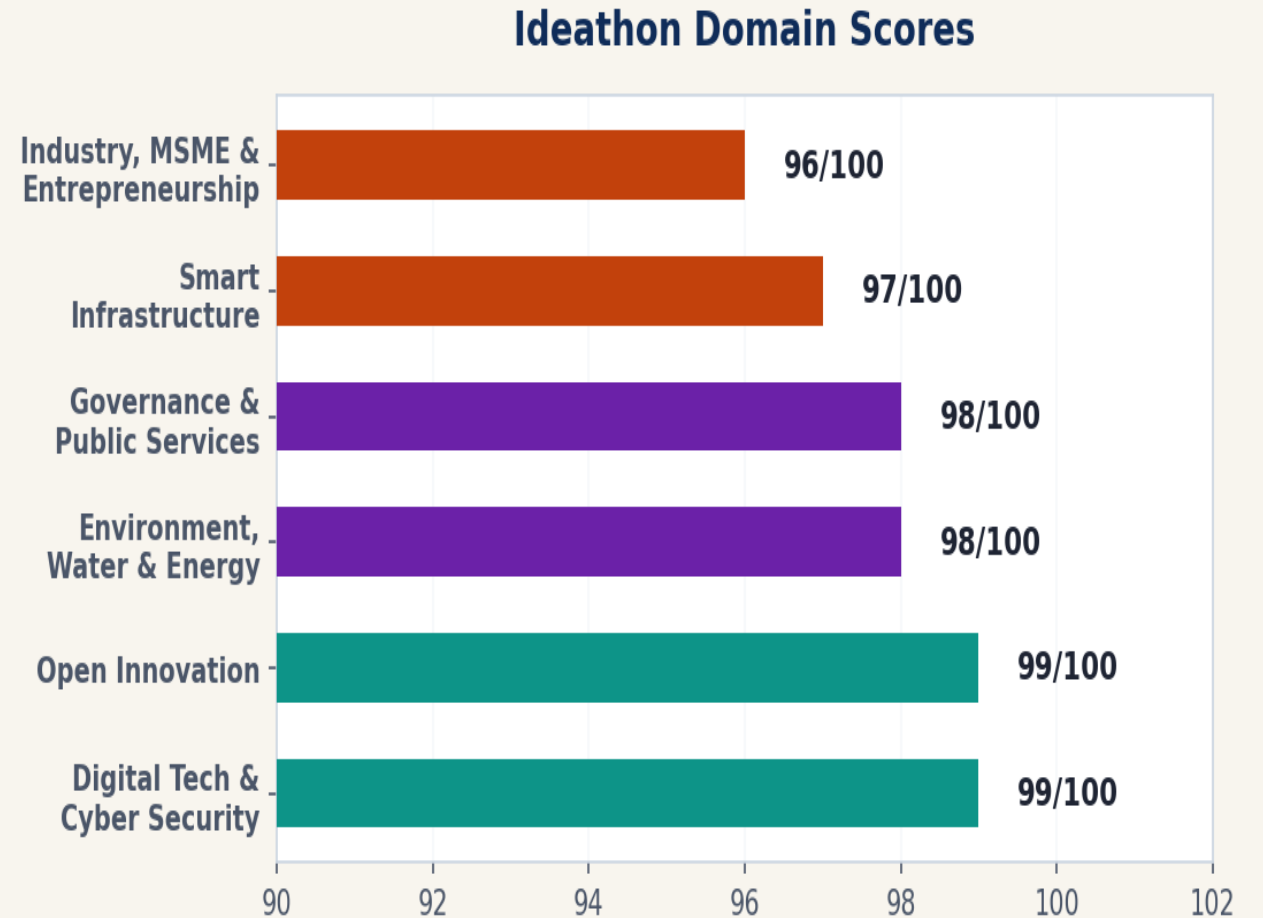
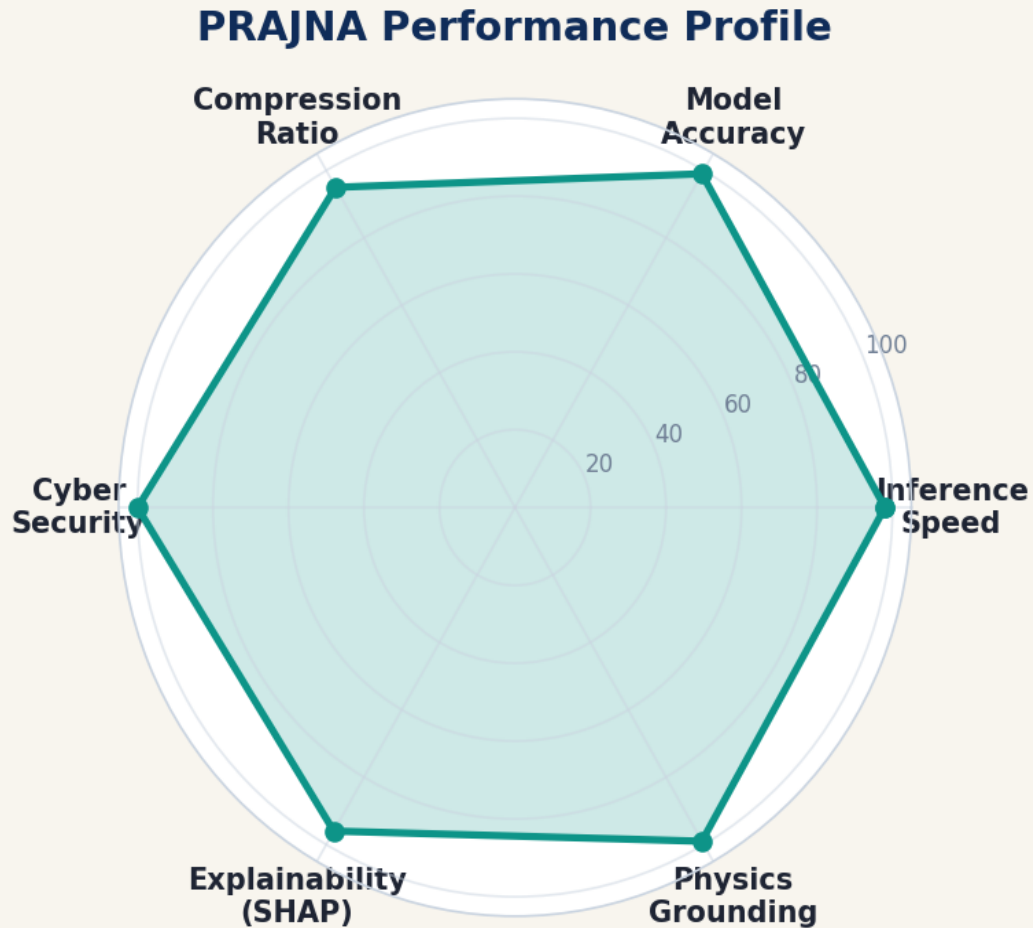
Pipeline unit tests passing



Note: Edge model shows 99.0% EOP accuracy; the larger foundation "teacher" model reaches 100% on the same set.

PERFORMANCE PROFILE

Multi-dimensional capability assessment across all critical metrics



IMPACT & SCALABILITY

Built for nuclear-grade stakes, designed to generalize

WHO BENEFITS

- Plant operators get earlier, physics-verified warning of fast-developing transients
- Regulators gain a tamper-evident advisory layer that logs and explains every alert
- Nearby communities benefit indirectly from an added, independent safety check

BEYOND NUCLEAR

The same physics-informed-model + zero-actuation-advisory pattern applies to any high-consequence cyber-physical system:



Power
Grids



Chemical
Plants



Water
Treatment



Industrial
MSMEs

HOW TO QUANTIFY IMPACT

Once piloted, impact is measurable as: seconds of extra operator lead-time before a threshold breach, number of sensor-tampering events caught, and reduction in false-alarm rates vs. legacy monitors.

STATUS & ROADMAP

12 months in — a working system, honestly still in progress

BUILT & VERIFIED

- ✓ 264.7M-parameter foundation PINN checkpoint
- ✓ 30,577-parameter distilled INT8 edge model (ONNX)
- ✓ Native JS inference runtime for the edge model
- ✓ 120 FPS 3D digital-twin dashboard with live SHAP overlays
- ✓ Automated test suite — 6/6 passing

NEXT MILESTONES

- 
- ▶ Hardware-in-the-loop (HIL) testing against a physical simulator
 - ▶ AERB software qualification pathway (AERB/SG/D-25)
 - ▶ Identify and engage a candidate pilot site — e.g. a PHWR facility in Haryana

Direct deployment capability for Haryana's critical energy infrastructure

Nuclear Protection

Gorakhpur Haryana Anu Vidyut Pariyojana (GHAVP)
4 × 700 MW PHWRs — ₹20,000+ Crore project

Grid Stability

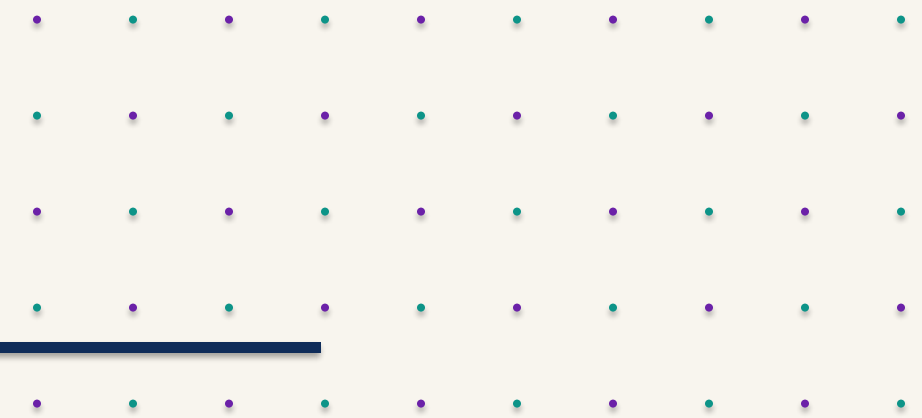
MSME Industrial Belts: Gurugram, Faridabad,
Panipat, Sonipat — 47 KB edge model on sub-stations

Zero-Trust Cyber

100% read-only hardware diode architecture
protects critical energy infrastructure from attacks

IN Global Leadership

Developed at DCRUST, Murthal — positions Haryana
as AI & Energy Security pioneer under Mission 2047



Thank You

PRAJNA — physics-grounded safety advisory for critical infrastructure

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